

A Complete Möbius and Morse–Bott Atlas for a Symmetric Matrix Riccati Flow

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Abstract

For every finite dimension we solve $\dot{X} = I - X^2$ on real symmetric matrices and classify its maximal classical chart, every signed-time pole, all forward limits, and the exact convergence remainder. We also prove a strict trace-gradient law, identify every involution component as a Grassmann Morse–Bott stratum with complete index, and derive the full Loewner formula for the nonlinear time map.

1 Exact flow and maximal chart

Let $X_0 \in \text{Sym}(n, \mathbb{R})$ and consider

$$\dot{X} = I - X^2, \quad X(0) = X_0. \quad (1)$$

Put $C = \cosh t$, $S = \sinh t$.

Theorem 1 (Möbius, pole, and limit atlas). *On the connected component through zero of $\{t : \det(CI + SX_0) \neq 0\}$, the unique classical solution is*

$$F_t(X_0) = (CX_0 + SI)(CI + SX_0)^{-1}. \quad (2)$$

If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of X_0 , the only finite chart singularities are

$$t_j = \text{arctanh}(-1/\lambda_j), \quad |\lambda_j| > 1, \quad (3)$$

with their spectral multiplicities. Thus the flow is forward global iff $\lambda_{\min}(X_0) \geq -1$. In that case

$$F_t(X_0) \longrightarrow I - 2P_{\ker(X_0 + I)}. \quad (4)$$

For each $\lambda > -1$ the exact scalar remainder is

$$f_t(\lambda) - 1 = \frac{2(\lambda - 1)e^{-2t}}{(1 + \lambda) + (1 - \lambda)e^{-2t}}. \quad (5)$$

Moreover $F_t \circ F_s = F_{t+s}$ whenever both expressions remain in one classical chart.

Proof. Solve the symmetric block-linear system

$$\dot{U} = V, \quad \dot{V} = U, \quad U(0) = I, \quad V(0) = X_0.$$

It gives $U = CI + SX_0$ and $V = SI + CX_0$. While U is invertible, $X = VU^{-1}$ obeys (1), proving (2); local uniqueness then gives maximality in that chart. The spectral theorem reduces F_t to $f_t(\lambda) = (C\lambda + S)/(C + S\lambda)$. Its denominator vanishes exactly as in (3). For $t > 0$ this happens precisely when $\lambda < -1$. The values -1 and 1 are fixed, every $\lambda > -1$ tends to 1 , and elementary exponential algebra yields (5). Multiplication of the two scalar Möbius matrices gives the addition law, hence the semigroup identity by functional calculus. $\square$

2 Gradient geometry and the complete equilibrium index

Define

$$\Phi(X) = \text{tr}(X^3/3 - X). \quad (6)$$

The Frobenius gradient is $X^2 - I$, so along (1)

$$\dot{\Phi} = -\|I - X^2\|_F^2. \quad (7)$$

In particular, continuity plus strict decrease rules out every nonconstant recurrent classical trajectory.

Theorem 2 (Morse–Bott involution atlas). *The equilibria are exactly the symmetric involutions. The component with p eigenvalues $+1$ and $q = n - p$ eigenvalues -1 is*

$$\mathcal{E}_{p,q} \simeq O(n)/(O(p) \times O(q)).$$

At every point of this component, the stable, unstable, and center dimensions of (1) are respectively

$$\frac{p(p+1)}{2}, \quad \frac{q(q+1)}{2}, \quad pq. \quad (8)$$

The center space equals $T\mathcal{E}_{p,q}$, so the critical manifold of Φ is Morse–Bott.

Proof. An equilibrium satisfies $S^2 = I$ and is orthogonally conjugate to $\text{diag}(I_p, -I_q)$, which proves the homogeneous-space claim. The linearization is $H \mapsto -(SH + HS)$. Writing a symmetric perturbation in plus/minus blocks gives eigenvalue -2 on the symmetric p block, $+2$ on the symmetric q block, and zero on the $p \times q$ off-diagonal block. Their dimensions are (8); infinitesimal orthogonal conjugacy identifies the last block with the tangent space. $\square$

Theorem 3 (Full derivative of the time map). *In an eigenbasis of X_0 , on every regular chart,*

$$(DF_t(X_0)H)_{ij} = \frac{H_{ij}}{(C + \lambda_i S)(C + \lambda_j S)}. \quad (9)$$

Proof. For $i \neq j$, the divided difference of f_t is $(f_t(x) - f_t(y))/(x - y) = ((C + xS)(C + yS))^{-1}$ because $C^2 - S^2 = 1$. Its diagonal limit is $f'_t(x) = (C + xS)^{-2}$. The standard spectral divided-difference formula, also valid across repeated eigenvalues by this limit, gives (9). $\square$

Source lineage

The citation supplies matrix-Riccati and symmetric-flow context, not a literature-priority claim for this packaging.

References

- [1] U. Helmke, “Isospectral flows on symmetric matrices and the Riccati equation,” *Systems & Control Letters* 16 (1991), 159–165. DOI: [10.1016/0167-6911\(91\)90044-F](https://doi.org/10.1016/0167-6911(91)90044-F).